

Alexander Marshall | Physicist

1900 Glenacre Drive – Kernersville, North Carolina 27284

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🌐 philosophiaephysica.org • Royster Doctoral Fellow at UNC Chapel Hill

HR Department

Corporation

123 Pleasant Lane

12345 City, State

January 27, 2026

To Whom it May Concern,

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Sincerely,

Alexander Marshall

Attached: Curriculum Vitae

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Education

Doctorate of Philosophy in Physics

The University of North Carolina at Chapel Hill

Royster Doctoral Fellow | Exams Passed | All But Dissertation

Chapel Hill, NC

In Progress

Master of Science in Physics

The University of North Carolina at Chapel Hill

Completed *en route* to PhD

Chapel Hill, NC

2025

B.S. in Physics, B.A. in Philosophy, Minor in Math.

Wake Forest University

GPA 3.91 | *Summa Cum Laude* | Stamps Scholar

Winston-Salem, NC

2022

High School Diploma

West Forsyth High School

Junior Marshall | Top 1% of graduating class

Clemmons, NC

2018

Academic Projects

Honors Thesis

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator, implementing the correction method of Dholakia et al. (2017) Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Masters Thesis

Physiological Strains in Poroelastic Media Induce Asymmetric Transport.

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Doctoral Thesis

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Additional Research.....

Chalcogenide Crystals for Quantum Computing **Winston-Salem, NC**
Wake Forest Nanotech Center, Dr. David Carroll 2021

Neurochemical Addiction Mechanisms In the Nocus Accumbens **Winston-Salem, NC**
Wake Forest School of Medicine, Dr. Evgeny Budygin 2019-2021

Cultural Impact of Oenology in the Peleponnese **Winston-Salem, NC**
Wake Forest Department of Anthropology, Dr. Karen Friederic 2019

Cancer and Postoperative Pain **Winston-Salem, NC**
Wake Forest Baptist Medical Center, Dr. Christopher Peters 2018

Scholarships and Grants

2022-2027: Royster Doctoral Fellowship. Five-year fellowship awarded for academic merit.

2018-2022: Stamps Scholarship. Four-year scholarship awarded for academic merit.

2019: Wake Forest Scholars Travel Grant. Funded summer anthropology project abroad.

Publications

Technical skills

Advanced Coding: MATLAB, LabVIEW, Python, $\LaTeX$, Git

Scripting: μ Manager, ImageJ

Data Analysis: Particle Tracking, Image Analysis, Stochastic Processes

Other: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Machine Vision, Atomic Force Microscopy, general lab skills, hazardous waste management

Presentations and Symposia

Generating a Three-Dimensional Lattice of Photonic Spheres **Winston-Salem, NC**
Symposium, Wake Forest University Physics Department May 2022

Diffusion in Strained Poroelastic Media **Chapel Hill, NC**
Thesis Proposal, UNC Chapel Hill Physics Department September 16, 2025

Strain Alters Anomalous Diffusion in Poroelastic Media. **Elon, NC**
Symposium, Elon University Physics Department October 1, 2025

Physiological Strains in Poroelastic Media Induce Asymmetric Transport. **Chapel Hill, NC**
Master's Defense, UNC Chapel Hill Physics Department November 17, 2025

Conferences

Undergraduate Research Day <i>Symposium, Wake Forest University URECA Center</i>	Winston-Salem, NC <i>September 2019</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, Duke University</i>	Durham, NC <i>September 15, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences</i>	Raleigh, NC <i>September 23, 2024</i>
Probing Structural and Force Dynamics during Phagocytosis <i>Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences</i>	Raleigh, NC <i>September 25, 2023</i>
Various topics <i>Annual Colloquium, UNC Applied Physical Sciences Department</i>	Chapel Hill, NC <i>2022-2025</i>
Strain Driven Transport In Poroelastic Media <i>8th International Soft Matter Conference, Soft Matter Association of the Americas</i>	Raleigh, NC <i>August 2024</i>
Strain Driven Transport In Poroelastic Media <i>Triangle Soft Matter Workshop, Duke University Physics Department</i>	Durham, NC <i>May 12, 2025</i>
Strain Driven Transport In Poroelastic Media <i>13th Carolina Biophysics Symposium, UNC Chapel Hill</i>	Chapel Hill, NC <i>November 6, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Royster Global Conference, Royster Society of Fellows</i>	Chapel Hill, NC <i>October 21, 2025</i>
Strain Driven Transport In Poroelastic Media <i>Biophysical Society Meeting, Biophysical Society</i>	San Francisco, CA <i>February 21, 2026</i>

Teaching and Mentorship

2021-Present: Many semesters experience as a TA.

2018-Present: Private tutor in advanced STEM fields.

Undergraduate Research Mentorship: 2023-Present

Outreach

Wake Forest Baptist Medical Center <i>Volunteer, over 400 accumulated hours</i>	Winston-Salem <i>2015–2017</i>
UNC APS Department <i>Open Flexure Microscope and Molecular Stamper</i> Constructing low cost microscopes for use in low-resource environments. Project has received multiple awards, including the UNC MakerFest Entrepreneurship Award.	Chapel Hill <i>2023-Present</i>
UNC Chapel Hill <i>UNC ScienceFest</i>	Chapel Hill <i>2023-Present</i>

Select Coursework

Graduate Mathematical Methods	2022
Graduate Quantum Mechanics	2022
Graduate Classical Dynamics	2022

Graduate Electromagnetism	2023
Graduate Statistical Mechanics	2023
Optics	2024
Fluid Dynamics	2025
Biophysics	2025
Partial Differential Equations	2022
Quantum Computing	2022

References

Richard Superfine, PhD: superfine@unc.edu	<i>Doctoral Advisor</i>
Kerry Bloom, PhD: kerry_bloom@unc.edu	<i>Reasearch Collaborator</i>
David Hill, PhD: dbhill@email.unc.edu	<i>Doctoral Committee Member</i>
Keith Bonin, PhD: bonin@wfu.edu	<i>Undergraduate Advisor</i>
Evgeny Budygin, PhD, PharmD: budygin@gmail.com	<i>Research Collaborator</i>

Languages

English: Fluent	<i>Native Speaker</i>
Modern Greek: Intermediate	<i>Conversational</i>
Classical Latin: Intermediate	<i>Several years of formal instruction</i>

Personal Interests

-Music	-Philosophy
-Running	-Backpacking